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Revolutionizing Well Interventions in Slim 3-7/8" Openhole, Global Firsts in High-Definition Spectroscopy and High-Pressure Abrasive Jetting Pin-Point Acid Stimulation Technologies on Real-Time Hybrid Fiber Optics Coiled-Tubing Platform

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Abstract

North Kuwait Jurassic Gas asset produces from Jurassic deep sour gas condensate carbonate reservoirs. High depletion causes differential pressure drilling challenges across the main producing multilayer reservoir. Increasing number of new wells have to be completed as slim 3-7/8" openhole in the main potential flow zones. Hence, intervention for formation evaluation logging and proper stimulation placement became challenging leading to inefficient stimulation that impacts the well production potential. This paper will demonstrate the design, execution challenges and successful results of revolutionary first global well intervention in such ultra-slim 3-7/8" openhole wells, integrating high-Definition Spectroscopy for formation evaluation and AbrasiJET Pin-point acid stimulation technologies deployed on Real-time Hybrid Fiber Optics CoiledTubing (CT) platform.

The Jurassic gas asset produces mainly from deep high temperature and initially high pressure tight complex heterogeneous carbonate reservoirs. Hence, well productivity potential significantly depends on formation evaluation logs and the effectiveness of subsequent selective stimulation treatments.

As a result of pressure depletion, the main producing reservoir experiences high differential pressure among its multilayers specially the bottommost major potential flow-zone where it becomes very challenging to land with same 6" openhole section as standard drilling design. Alternatively drilling 3-7/8" deepening slim barefoot is unavoidable despite utilizing Managed-Pressure Drilling system. This challenge has implied significant limitation to running formation evaluation logs and efficient selective stimulation of the 3-7/8" slim openhole section due to limitation on conventional Coiled-tubing intervention capabilities. To overcome such intervention challenges, the integrated asset's field development and operation teams has conducted technical feasibility and intervention risk management study on running High-Definition Spectroscopy logging for formation evaluation and High-Pressure Abrasive Jetting Tool (HPAJT)pin-point stimulation deployed on Real-time Hybrid Fiber Optics CT platform in ultra-slim 3-7/8" openhole. This revolutionary integrated intervention technique has been implemented successfully as the first global

deployment in two Jurassic gas wells late 2024. The success of acquiring formation evaluation logs has highly improved stimulation design through better distribution and efficiency across the openhole targeting low perm intervals from Logs, for the subsequent efficient pin-point stimulation placement.

The initial production tests from both wells showed outstanding results. The first well was treated with 27 pin-point stimulation stations and achieved 50% additional production gain exceeding the production forecast. The second well had initially been acid stimulated with bull-heading techniques where it could not be activated, however initial production exceeded its production forecast post executing the pin-point acid stimulation.

This groundbreaking Technology combination and success sets a new benchmark for well intervention and establishing another unparalleled standard in Jurassic ultra-deep well intervention operations. This success opens new opportunities to conduct similar treatments in analog cases to maximize production potential of such slim openhole wells.

Introduction to North Kuwait Jurassic Reservoirs

The Kuwait North Gas Field asset includes carbonate reservoirs with near critical gas and volatile oil fluids. These reservoirs are complex in terms of reservoir quality and fluid and rock properties, and are very challenging with regards to production both upstream and downstream with typical depths from 14,000 to 18,000 ft MD. The produced fluid is sour and contains high H₂S (up to 15%) and CO₂ (up to 6%). The reservoir pressure ranges from depleted 5,000 psia to almost virgin 12,000 psia with an average reservoir temperature of 275°F.

The recovery from these reservoirs is uncertain if a conventional development strategy is applied. Therefore, a dedicated full development plan applying integrated upstream and downstream technologies is critically important.

The main pay zone is Middle Marrat reservoir which is a tight limestone (Calcite > 90%) with very low porosity and permeability, streaked with layers of dolomite that have superior rock properties. A pressure transient analysis (PTA) that was performed in various wells showed dual porosity behavior with low matrix permeability as storage volume. The average matrix permeability is less than 1.0 mD (Al-Saeed et al 2017).

Due to the high Jurassic carbonate reservoir tightness, permeability contrast among different flow units and dual permeability effect (matrix and natural fractures), the well's productivity potential significantly depends on the effectiveness of subsequent stimulation treatments of such complex heterogeneous reservoirs to improve productivity and connect the natural fractures when present. [Figure 1](#) and [Figure 2](#) show the Kuwait Jurassic Stratigraphy and typical log showing high permeability contrast among reservoir layers, respectively.

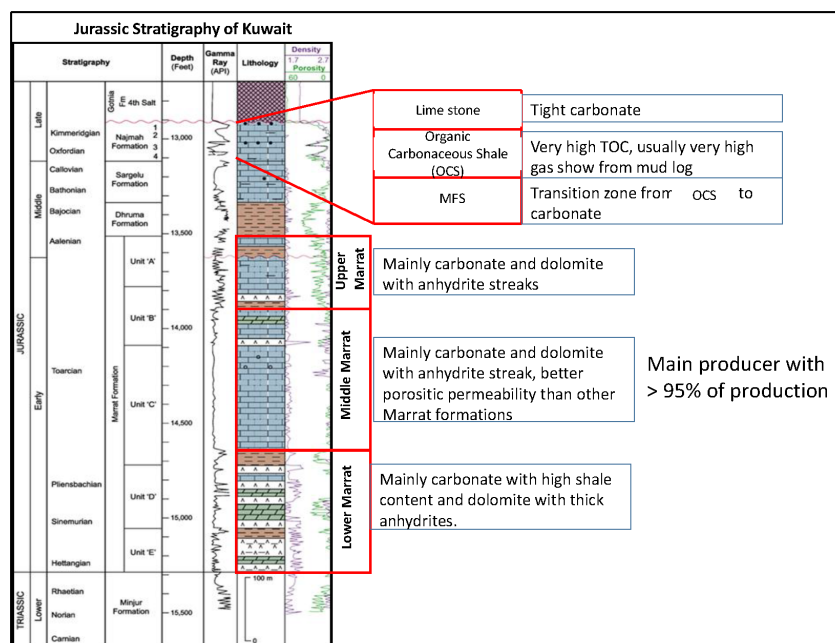


Figure 1—Kuwait Jurassic Stratigraphy

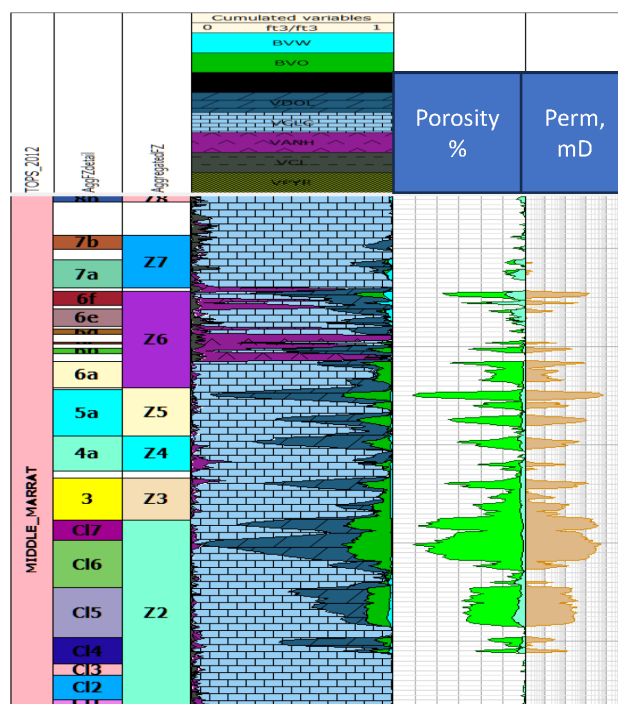


Figure 2—Typical log showing high permeability contrast among reservoir layers

The Middle Murrat multilayers (flow zones) reservoir is historically the main producer reservoir in North Kuwait Jurassic fields since the early exploration phase where it contributes more than 90% of total fields production. Therefore, it experienced high depletion compared to its original reservoir pressure as it produces under depletion mechanism with no pressure support.

Jurassic Gas Completion History

Selecting proper completion design is crucial to overcome reservoir challenges and ensure efficient and effective acid stimulation treatments (matrix and fracturing) of these tight heterogenous formations. This

required a convenient diversion mechanism during the stimulation to enhance the productivity of each individual reservoir sub-flow unit and enable selective future flexibility of re-stimulation and reservoir management. Early field development considered cementing a 5-in liner across the reservoir section with a 3.5-in upper tubing completion (Figure 3), but it had significant limitations in-terms of rig-less selective stimulation. Alternatively, the completion strategy was shifted to a monobore completion consisting of a 4.5-in cemented liner across the reservoir section and 4.5-in upper tubing to enable rig-less plug and perf selective stimulation of reservoir layers from bottom up (Figure 4).

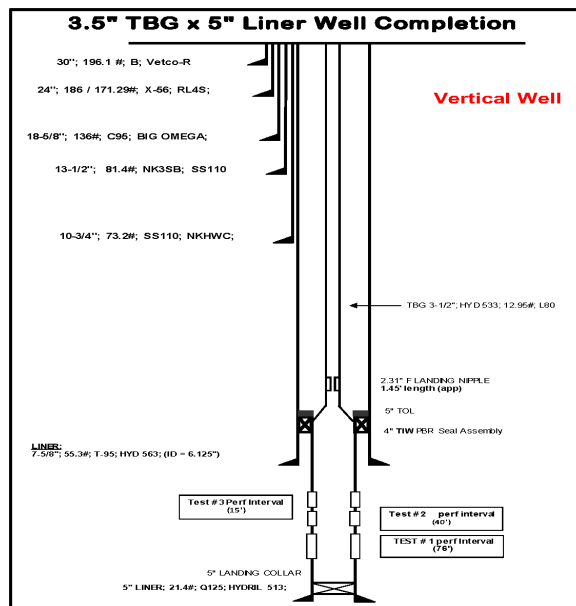


Figure 3—shows a typical Jurassic well of 3.5" / 4.5" tubing and 5" cemented liner tapered completion schematic example.

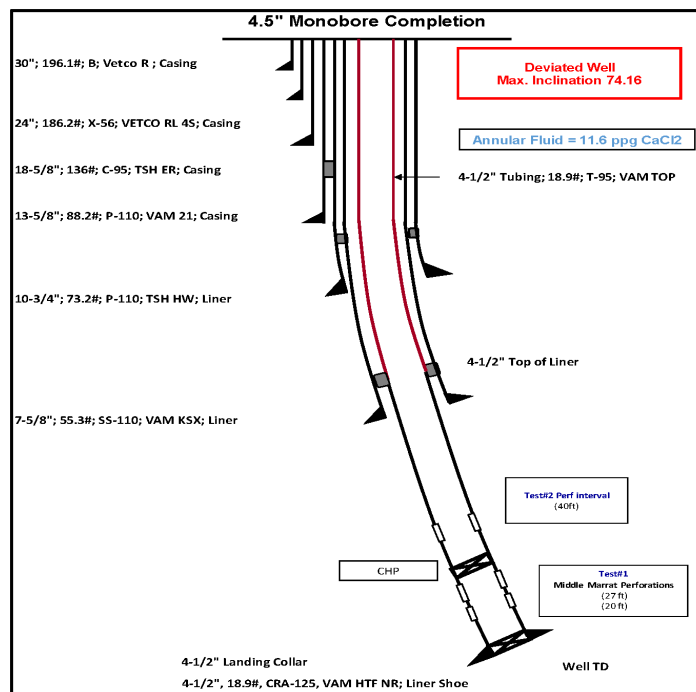


Figure 4—shows a typical Jurassic well of 4.5-in monobore cemented liner completion schematic example.

Monobore 4.5-in cemented completions showed some accessibility limitations specifically in highly deviated wells (>60 degrees) where wellbore cleaning challenges exist and could be considered as sub-optimal due to the unavailability of a slim drilling string to clean inside 4.5-in liner. This issue has adversely affected some wells perforation and stimulation intervention plans which in turn delayed well delivery to production. Furthermore, cemented monobore completions could limit connectivity to the reservoir's natural fractures, which affects the overall productivity potential, and requires multiple well interventions for perforating long intervals (up to 650 ft net perforations) and subsequent isolating of planned stimulation stages (on average five stages per well). This can add high operational challenges that have the potential to consume additional resources and result in mobilization logistics taking longer to deliver new wells to production.

Consequently, the North Kuwait Jurassic gas team has adjusted the completion strategy to open hole HPHT Multi-Stage ball drop Completions (MSC) to overcome such reservoir complexity (Figure 5). By making this modification, many challenges of cemented monobore completions have been eliminated or minimized such as the need for monobore completion's wellbore cleaning, avoided multiple perforation intervention operations challenges and risks, eliminated cement quality risks and uncertainties, and improved overall cost. These updates ultimately fast tracked well delivery to production to meet the asset production target by significantly reducing operation time from approximately one month when using a plug and perf technique of monobore completions, to less than one week with continuous and less subsurface intervention operations.

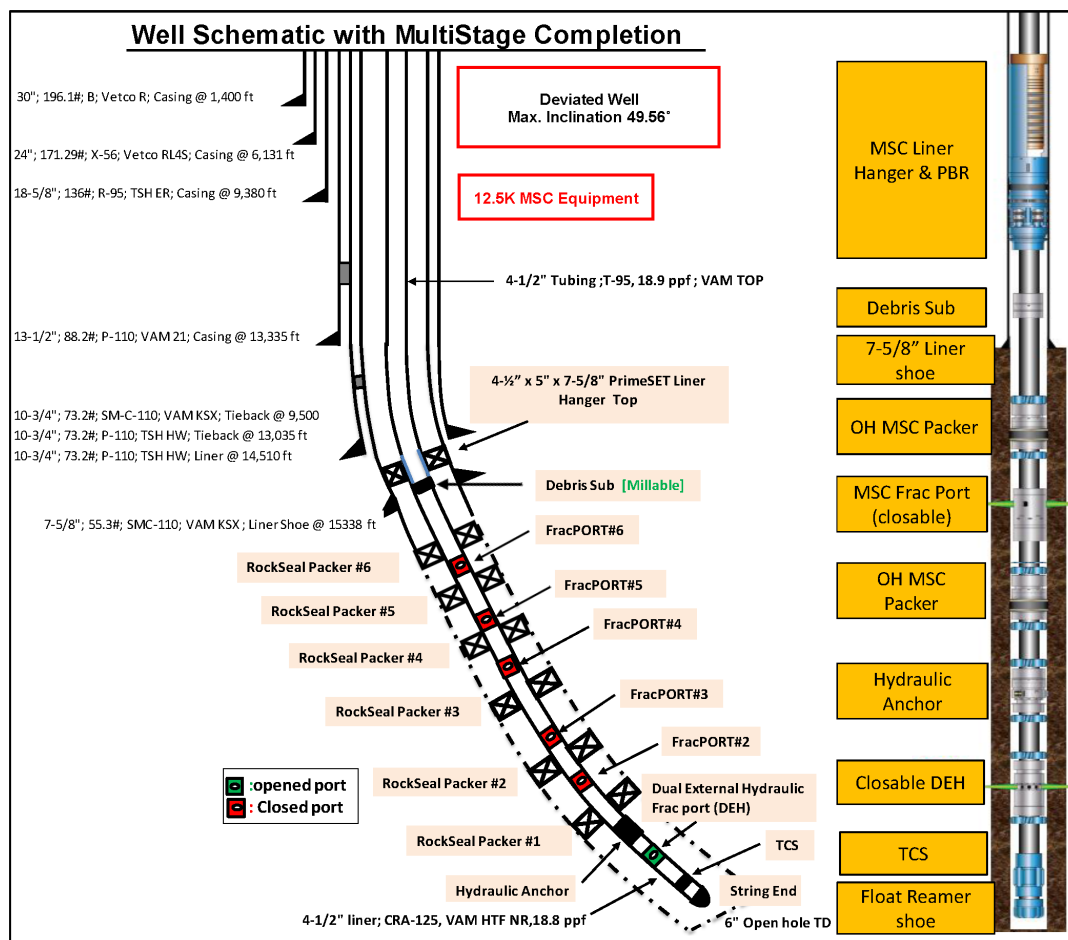


Figure 5—Lower MSC showing main components of completion string and well MSC schematic example.

Middle Marrat Depletion Effect on Lower Completion and Associated Challenges

The Middle Marrat multilayers (flow zones) reservoir is historically the main producer reservoir in North Kuwait Jurassic fields since the early exploration phase where it contributes more than 90% of total fields production. Therefore, it experiences high depletion from its original reservoir pressure as it produces under depletion mechanism with no pressure support.

The Middle Marrat reservoir used to be drilled in single 6in openhole (OH) section down to its bottom most flow Zone (reservoir layer) which is the main production flow zone and used to be opened only for production for almost the first 10 years of NK JG fields early production. Then, this OH section to be completed with 4.5in cemented liner or 4.5in Openhole MSC to enable later selective stimulation and testing. As a consequence, these bottom flow zones started to experience higher reservoir pressure depletion compared to the upper flow zones / reservoir layers. As a result, this has imposed high differential pressure across the Middle Marrat reservoir layers that makes it challenging to complete drilling the full reservoir within the single 6in openhole section. Therefore, the end of the 6in section had to be called off during drilling operations at shallower depth in some well locations due to high risk of differential sticking. To be able to drill to the bottom high potential flow zones, the drilled 6in OH section is completed with cemented 5in liner then the remaining reservoir section is drilled and completed as slim 3-7/8in Openhole (Figure 7). The new resulted slim 3-7/8in OH plus 5in cemented liner in the Middle Marrat reservoir section imposed new stimulation and testing challenges. These challenges have implied significant limitations to running formation evaluation logs and efficient selective stimulation of the 3-7/8" slim openhole section due to limitation on conventional Coiled-tubing intervention capabilities due to associated challenges and risk of getting stuck specially with unavailability of required special fishing tools for such size.

Starting 2020, Increasing number of new wells have to be completed as slim 3-7/8" openhole in the main potential flow zones. Hence, intervention for formation evaluation logging and proper stimulation placement became challenging and have to bullhead the acid stimulation treatment from surface across the full long slim 3-7/8in OH section (up to 1000 ft in some wells) without any control other than fluid chemical diversions which led to inefficient stimulation that impacts the well production potential. Therefore, it was critical to enable tools intervention inside the slim 3-7/8in OH to run formation evaluation logs to help evaluating the reservoir quality, potential to support stimulation design then to pump the stimulation treatment through coli tubing pin-point-stimulation to improve the stimulation efficiency.

Solution

Push-back was the typical decision for openhole access of wells completed with 3-7/8" slim OH in Middle Marrat formation, envisaging the intervention high challenges to run normal logging and stimulation tools. Correlation with offset wells was the only approach for formation evaluation, and only stimulation through bullheading was allowed. This implied high uncertainty on formation evaluation potential and stimulation design for the 3-7/8" open-hole section.

Following technical discussions and risk mitigation analysis, an integrated multi-team collaboration succeeded implementing Real-time special Coiled-tubing Unit (CTU) intervention for the First time Worldwide utilizing High-Definition Spectroscopy and AbrasiJET Pin-point Acid Stimulation Technologies on Real-time Hybrid Fiber Optics Coiled-tubing Platform. Traditional open-hole access concerns were perfectly managed with the telemetry real-time data.

Implementation of New Technologies Array

Combination and integration of technologies bundle enabled a complete solution of safe openhole accessing, evaluating and stimulating otherwise the completion type is inaccessible/ challenging. Technologies

involved in the well-1 solution offerings as illustrated and summarized per below (Figure 6). Considering the intervention challenges in 3-7/8" OH, the only way without this solution is to use the correlations from offset wells for formation evaluation, might be over/under estimations, and bullhead the stimulation with majority of acid is directed to high permeability intervals/ below liner shoe.

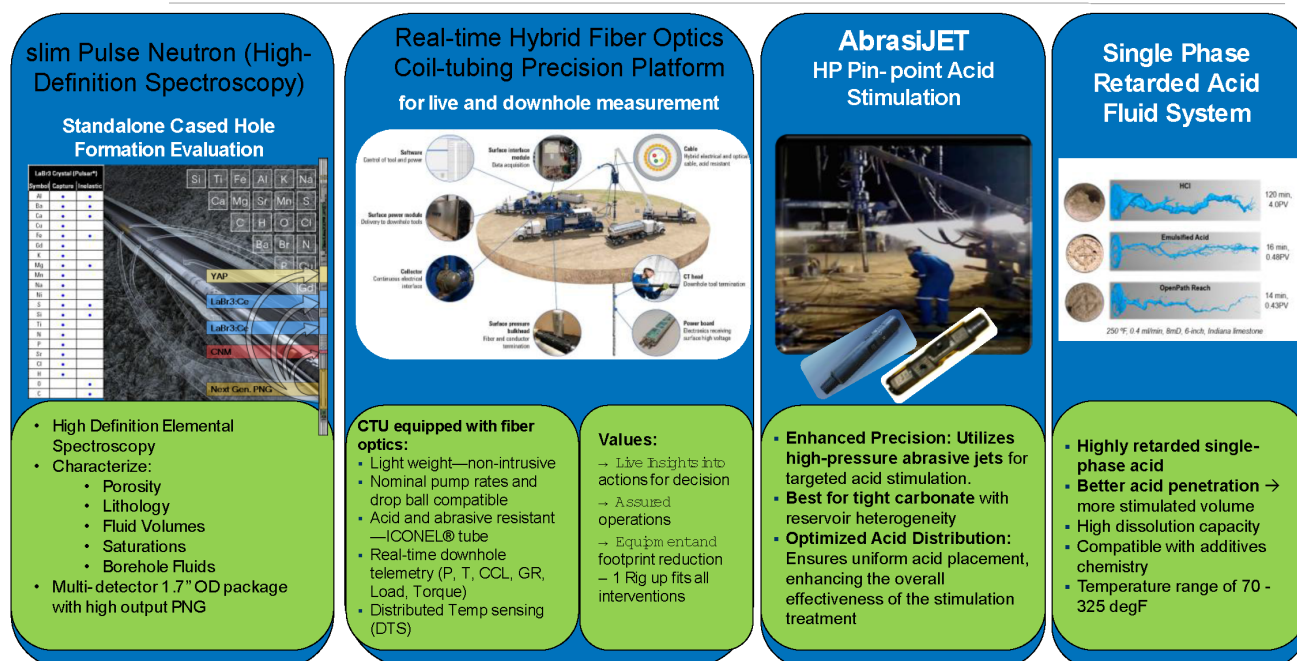


Figure 6—Well-1: Implemented New Technologies Summary

The technologies integration of the solution enabled optimization of stimulation by precise identification of damaged zones and Pin-point stimulation for treating the target zones.

Table 1 below listing and demonstrating the values enabled by adoption of the new technologies:

Table 1—values enabled by adoption of the new technologies

	New Practice: Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform which enable intervention in 3-7/8" OH	No Intervention in 3-7/8" OH
Formation Evaluation logs (e.g. Slim Pulse Neutron)	- Better formation potential evaluation - Better Stim planning. - Detection of Barite invasion in OH (on-spot Hi Vis pill circulation decision)	- Correlation from offsets (over/under estimation) - Use offsets data for Stim design. - No idea of Barite content/ invasion in OH
Stimulation Efficiency	Pin-point stim improves stim distribution & efficiency across the OH targeting low perm intervals from Logs	Only Bullheading where majority of acid goes to high perm interval, would be low efficiency

Case Study Well-1

Well-1 was drilled as Middle Marrat (MM) reservoir development well, passing through all MM Flow zones (FZ) from FZ8 to middle FZ2 with Middle Marrat being primary and Najmah secondary target. The well is deviated with 56°C degree inclination at TD 15,890 ft. MD (Figure 7).

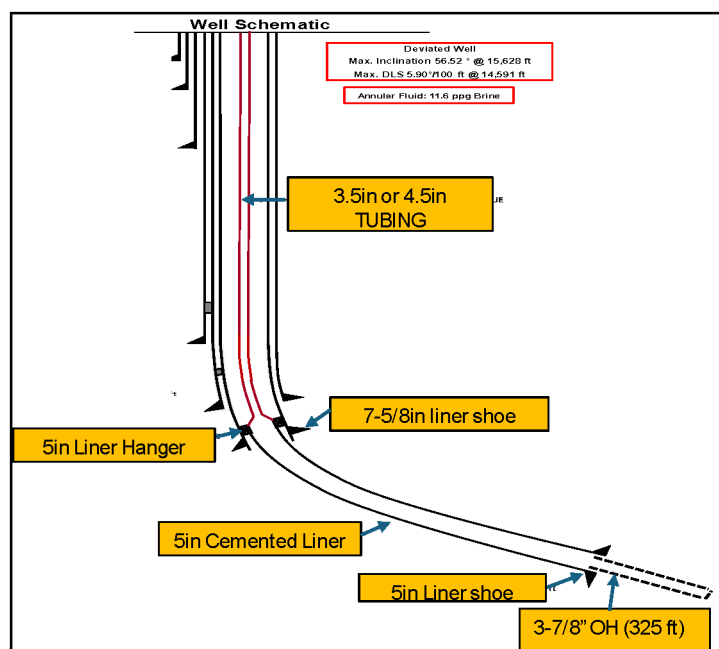


Figure 7—Well-1: Slim 3-7/8in OH × 5in cemented Liner Lower Completion.

Well-1 is placed in the southwest part of Raudhatain field, expected to be non-fractured area even though Coriolis analysis identified several possible fractured zones across primary target, additionally natural fractures were interpreted in a nearby offset. Due to the reservoir pressure depletion and drilling challenges to continue drilling 6in openhole to the planned Total Depth (TD), 5in liner was lowered and cemented to cover until middle FZ4, then 3-7/8" openhole deepening drilled crossing FZ4, FZ3 and well was TD in the middle of FZ2 with total 3-7/8in OH section of 325 ftMD.

The estimated production was from 2.0 to 6.0 MMSCFD based on offset wells' mixed production performance. Low reservoir pressure was estimated from offsets.

It was necessary to acquire OH logs in the 3-7/8" OH section and evaluate reservoir properties such as porosity, permeability, volumetrics (lithology) and water saturation to provide input for the stimulation program. The open-hole logging program designed to meet logging objective considered single run with slim Pulse Neutron (High-Definition Spectroscopy) logging tool deployed on Real-time Hybrid Fiber Optics Coiled-tubing Platform conveyance system.

Key Value Drivers and Candidate Screening Methodology

A significant improvement of stimulation efficacy is the key value driver reflected in the pilot project for implementation of the pinpoint Abrasijet stimulation in North Kuwait.

The starting point is the barefoot slim hole wellbore, the challenges and inability to stimulate it effectively by bull heading. The poor stimulation efficiency observed in previous slim hole barefoot wells was one of the key elements driving the implementation of pinpoint Abrasijet Stimulation in North Kuwait.

Candidate screening and Sweet Spotting

Several factors were considered when selecting the right candidate for the pilot implementation of the pinpoint Abrasijet stimulation in North Kuwait. The primary consideration is the Permeability and Permeability-Height function (KH). The pilot well selected must have a minimum average matrix permeability of 1.0 millidarcy (mD) across the selected intervals. Conversely, a cutoff for the Permeability-Height function (KH) was also implemented.

Another fundamental selection criterion is the current static reservoir pressure and the effect of depletion on the stimulation efficiency and recovery of spent fluids in the well bore. Several previous wells prior to

Well-1 with reservoir pressure depletion were typically stimulated by bullheaded stimulation. This resulted in huge losses during stimulation, poor stimulation effectiveness, poor clean up and ineffective recovery of spent fluids from the wellbore. A decision was taken to consider reservoir depletion for pinpoint Abrasijet stimulation deployed via Coiled tubing to reduce the acid stimulation fluid load, thereby aiding fluid recovery, clean-up and activation. Figure 8 below shows the key selection criteria for the sweet spots and the pinpoint Abrasijet stations

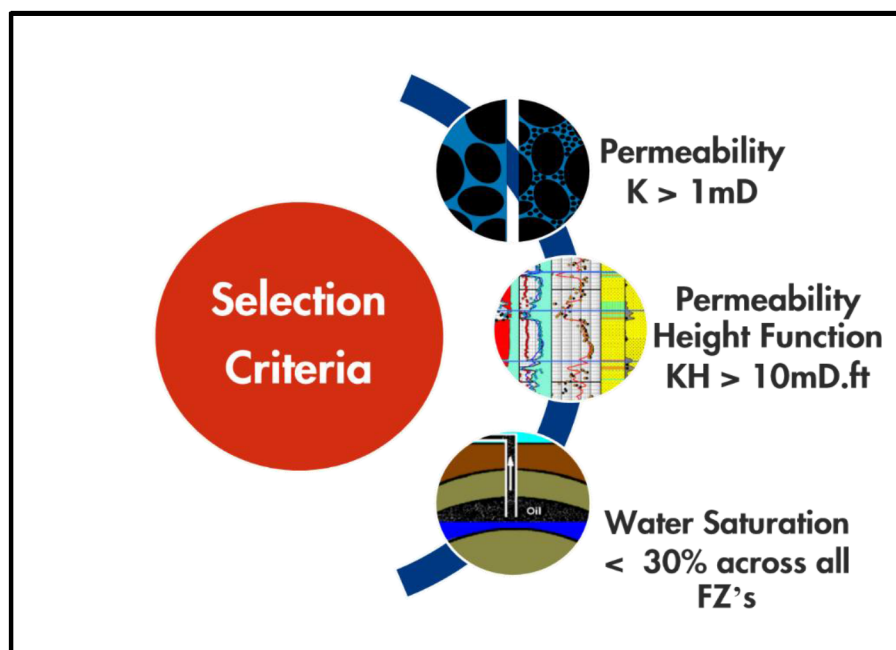


Figure 8—Key selection criteria for Candidate wells sweet spots

Most importantly, a crucial competent is wellbore access to TD in the slim hole open hole section. In Pilot Well-1, full-bore access to TD in the open hole was established prior to the logging evaluation. Ensuring the wellbore stability was important to mitigate the risk of getting stuck with the coiled tubing and Abrasijetting tool BHA during the stimulation.

A summary of the parameters considered when choosing the candidate wells are as follows:

- Matrix Permeability (K)
- Permeability-Height function (KH)
- Current reservoir pressure
- Water Saturation
- Reservoir depletion in the near wellbore region
- Current water production in offset wells across flowzones

Other criteria for selection to screen other candidates were:

- Access to TD in Open Hole section
- Internal diameter of the upper completion
- Expected gain after stimulation with pinpoint Abrasijet stimulation compared with typical bullheaded acid stimulation

- Nodal analysis evaluation: - Expected post stimulation skin value

Based on these criteria listed above, 27 number of sweet spots were identified in the slim hole section of candidate Well-1. It is important to reiterate that the sweet spots were identified using on the acquired PNx log, this was a key data point that enabled the selection of the right candidate and proper identification of the pinpoint sweet spots or stations.

High-Definition Spectroscopy Service (HDSS) logging

The HDSS tool (Rose et. al 2015) provides high-resolution formation evaluation through multi-physics Pulsed Neutron induced Gamma Ray spectroscopy in both cased and open hole environments. Its multi-detector design enables robust self-compensation and environmental correction, yielding high-quality porosity data via Thermal Neutron Capture Cross-Section (Σ) and Hydrogen Index (TPHI) measurements. The tool also captures an extended suite of elemental concentrations, enabling detailed mineralogy and fluid characterization. Error! Reference source not found. Figure 9 shows the HDSS tool architecture, elemental spectroscopy measurements and the Petrophysical outputs. A few unique measurements make the HDSS tool ideal to address formation evaluation challenges for the Kuwait Jurassic reservoirs:

- Multiple conveyance options with real time surface read out:
 - Conventional wireline – for low angle wells
 - Tractor conveyance – for high angle or horizontal cased wellbores
 - Hybrid Coil Tubing conveyance – for accessing and logging complex trajectories with barefoot sections
- Elemental concentration measurements of Iron, Chromium and Nickel from the near detectors provide an excellent source of characterizing near wellbore contributions from the completion during logging behind casing. This has been especially beneficial in the wells typically completed with chrome completions.
- Elemental concentration measurements of Barium, Sulphur, Potassium, Calcium and Chlorine from the near detectors provide an excellent source of characterizing near wellbore contributions from the borehole fluids. This has been especially beneficial in the wells typically drilled with Barite (BaSO_4) weighted oil-based mud (OBM). The heavy OBM is displaced with Calcium or Potassium Chloride completion brines during logging HDSS in some cases.
- Organic Carbon measurements related to live hydrocarbon or OBM can be used for volumetric fluid saturation analysis.
- Measurements of Fast Neutron Capture Cross-Section (FNXS) to quantify reservoir fluid phases. Fast Neutron interactions are very sensitive to the Atom Density of the rock & fluid being analyzed (Zhou et. al 2016). FNXS values of the tight rock matrix and rock matrix with liquid filled porosity are in the same range. A significant reduction in measured FNXS compared to the theoretical rock matrix FNXS is seen when the pore space is filled with light hydrocarbons or gas. Heavy liquids and solids, typically from the borehole mud, tend to increase measured FNXS values significantly.

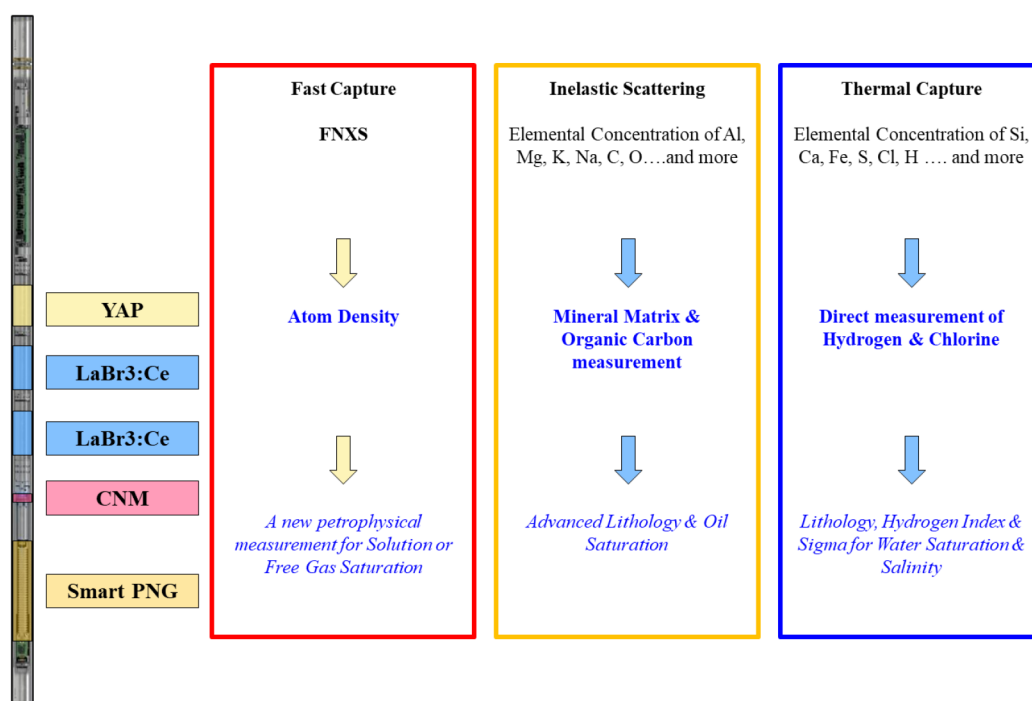


Figure 9—HDDSHardware Architecture, Measurements & Petrophysical Outputs

By utilizing the multi-physics measurements, one can not only perform a detailed formation evaluation, but also easily characterize the different fluids in the rock matrix and in the wellbore thereby delineating intervals with live hydrocarbons or invaded OBM filtrate (liquid phase) and OBM Barite (solids) invasion. Intervals with OBM solids invasion can be targeted for aggressive pin-point acid jetting for bypassing the skin and connecting to the reservoir.

Pin-Point Stimulation Treatment Design and Execution Summary

Abrasive Jetting Technology. The original design and purpose of the high-pressure abrasive jetting tool (HPAJT) is utilized to create slots by pumping sand or proppant slurry at high differential pressure across its nozzles as shown in in Figure 10. These nozzles create a high velocity stream estimated at 600-ft/sec that can perforate through tubulars, cement sheath, and formation rock. (Al-Atwi et al. 2012). The tool has 4 nozzles spaced at 90-deg where nozzles can be plugged using a plug assembly.



Figure 10—High-Pressure Abrasive Jetting Tool

The slot shape and dimensions are directly related to several factors as shown in Figure 11. These factors are the nozzle size, pumping rate, jetting time, nitrogen pumping rate, back pressure exerted against the jet, differential pressure across the nozzle, solid concentration of the slurry, and stand-off between the abrasive jetting tool and the casing or open hole. (Al-Atwi et al. 2012).

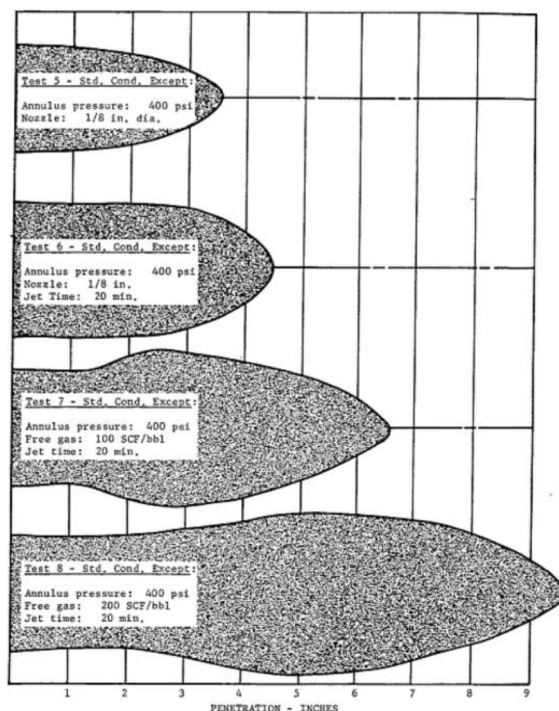


Figure 11—Slots shape and effect of jetting time, nozzle size, and nitrogen in open hole (Dowell test-1965)

As all of the factors mentioned above play an essential role in the tool deployment, job design and results. It is very important to understand the functionality of each factor for maximum tool efficiency.

Nozzle Size. The higher the number and size of nozzles, the higher the pumping rate needed to generate the required differential pressure across each nozzle. Since the job is conveyed with CT, the pumping rates are limited by the size of the CT string and the minimum inside diameter (ID) of the BHA. (Al-Atwi et al. 2012). To achieve the optimum penetration, the differential pressure at the nozzle needs to be from 2,200-psi to 2,600-psi. As the CT size used was 1.75-in the tool was designed to have 3 nozzles of 0.125-in and have a CT pumping rate of 1.6-BPM.

Jetting Time. Previous experiments show that the longer the jetting time the higher the deeper the penetration. Hence more, jetting more than 30 minutes is not recommended as additional penetration beyond this time is negligible. (Al-Atwi et al. 2012).

Pumping Rate. Pumping the fluid at a higher rate will result in deeper penetration and better results. However as the nozzles are tungsten carbide, higher pumping rates and longer jetting times, will result in eroding the nozzle faster which can result in making extra runs to change out the nozzles. (Al-Atwi et al. 2012). As mentioned above, the pumping rate is set for 1.6-BPM to achieve maximum penetration.

Drop Ball. As a multipurpose tool solution and to avoid having any mix up in hydraulics between operation of the abrasive cutter and the downhole nozzle, the abrasive cutter was designed to have a drop ball shifting sleeve that will allow isolating the downhole nozzle once CT reaches the target depth. The drop ball sleeve was designed to be shifted with a 7/16-in ball and will activate once the brass shear screws pressure exceeds 2,200-psi.

Single-Phase Retarded Inorganic Acid System: a Game-Changer in High-Temperature (HT) Carbonate Stimulation

Stimulating carbonate formations traditionally involves the use of chemicals such as hydrochloric acid (HCl) and emulsified acids. While effective, these methods have their limitations, including complications related

to viscosity and extended mixing times. This section introduces a groundbreaking solution - the Single Phase Retarded Inorganic Acid System (SPRIAS).

It revolutionizes carbonate stimulation by addressing the limitations of traditional methods. The engineered fluid combines the solubility of HCl acids with the retardation characteristics of emulsified acids, providing a unique solution for complex challenges. Unlike emulsified acids, SPRIAS offers comparable viscosity to HCl, reducing tube friction pressure and enabling faster pumping rates. Additionally, its ability to create deep wormholes, particularly in lower-pressure wells, positions SPRIAS as an optimal choice for enhancing stimulation in challenging wellbores.

Retarded acid systems are specifically engineered for carbonate formation acidizing, offering unique advantages. Their slow reaction rates allow deeper penetration with reduced acid volume, making them a superior choice to standard 15% HCl treatments. These advanced systems exhibit a higher solvent capacity per gallon of acid spent compared to organic acids. Remarkably, they demonstrate compatibility with various acid systems and additives, enhancing versatility and effectiveness.

Stability testing of the SPRIAS system under reservoir conditions has confirmed its efficiency in carbonate rock matrix treatments. Even after eight hours at formation temperature and room temperature conditions, the acid system remains stable, ensuring reliability during operations.

Historically, emulsified acids have been preferred in high-temperature carbonate reservoirs due to their slow reaction rates facilitating deep acid penetration. However, operational complexities, including extended emulsion mixing times and high tubing friction pressure losses, have driven the search for superior alternatives. SPRIAS, with its ability to create deep wormholes at low injection rates, eliminates these drawbacks and promises to revolutionize carbonate stimulation in the Middle East and beyond.

Figure 12 depicts a pore-volume-to-breakthrough (PVBT) experiment conducted at 250°F on a 1-in diameter $\times$ 6-in length Limestone core. The comparison involves 15% HCl and SPRIAS, both with similar HCl acid concentration, demonstrating that the PVBT of SPRIAS is significantly smaller than that of HCl.

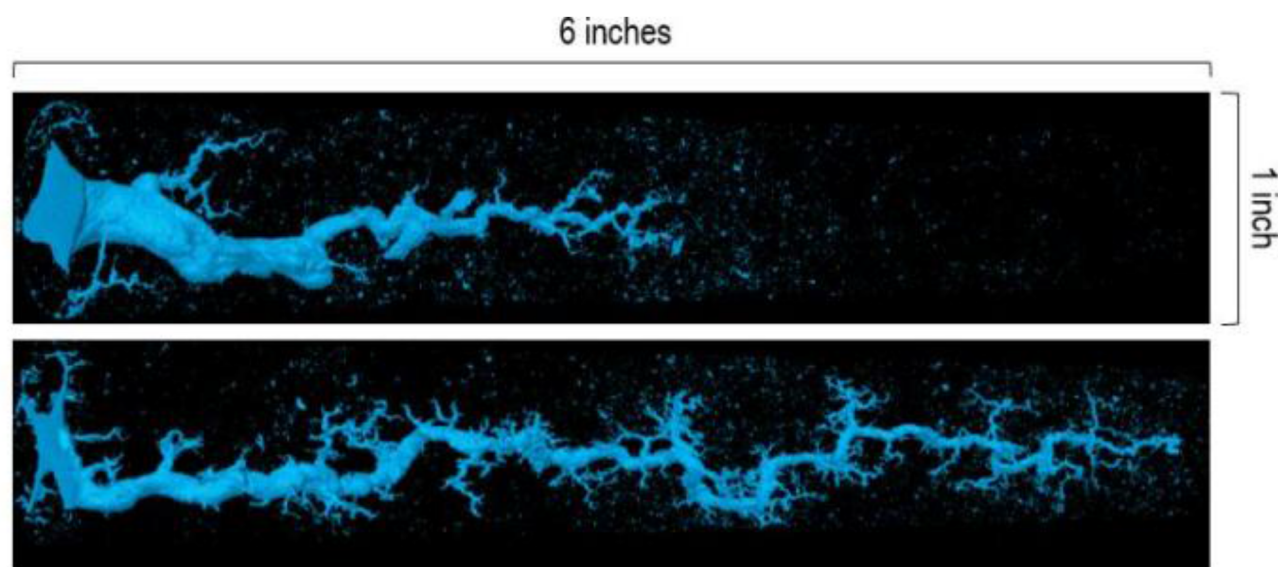


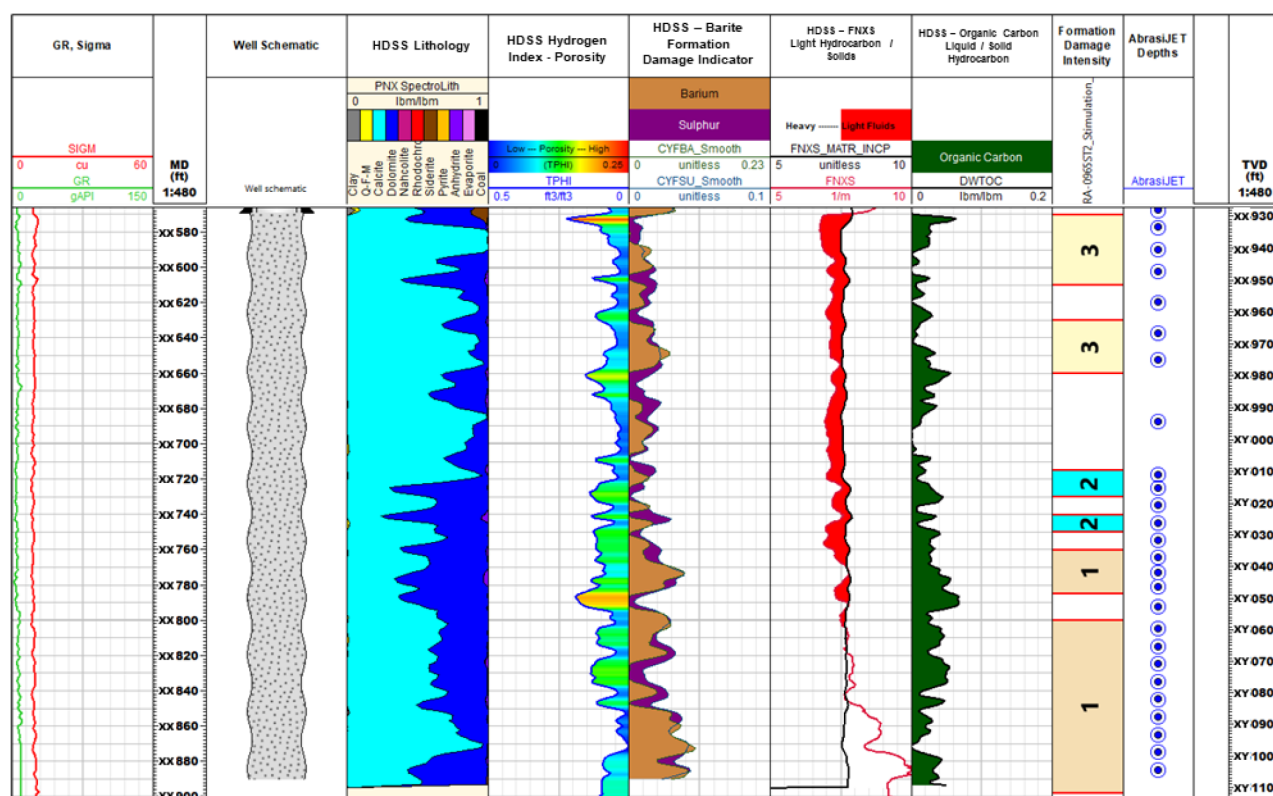
Figure 12—Comparative PVBT Experiment on a Limestone Core at 250°F and 0.4 ml/min Pumping Rate: 15% HCl and SPRIAS (SLB)

SPRIAS represents a significant advancement in carbonate stimulation technology. Its unique combination of solubility, retardation, and compatibility offers a promising solution to the challenges faced in traditional methods. As demonstrated through stability testing and successful field applications, SPRIAS is poised to enhance the efficiency and effectiveness of carbonate reservoir enhancement. Its adoption in various fields across Kuwait signifies its potential to transform carbonate stimulation practices worldwide.

Job Execution and Findings

The HDSS measurements provided invaluable information about the pay zones, fluid volumes and formation damage as listed below (Figure 13):

- Elemental yields of Calcium & Magnesium were used to determine the Carbonate lithology (Track 3) and Dolomitized intervals typically associated with porous pay zone in the Middle Marrat formation.
- Hydrogen Index (Track 4), FNXS (Track 6) & Organic Carbon Logs (Track 7) were used for formation evaluation and determine intervals with live hydrocarbon
- Barium & Sulphur Yields (Track 5) were used to identify intervals with the presence of Barite.
 - Barite invasion into the formation in many tight intervals with live hydrocarbon can be seen
 - An elevation in FNXS & Barite is seen below XX820ft. This is 130ft out of 325 ftMD open hole section. The elevation is interpreted as a column of stagnant solids from the mud and deep mud solids invasion.



Zone 1: Interval with excessive Barite due to Barite in the wellbore or deep invasion

Zone 2: Moderate Barite invasion

Zone 3: Low Barite invasion

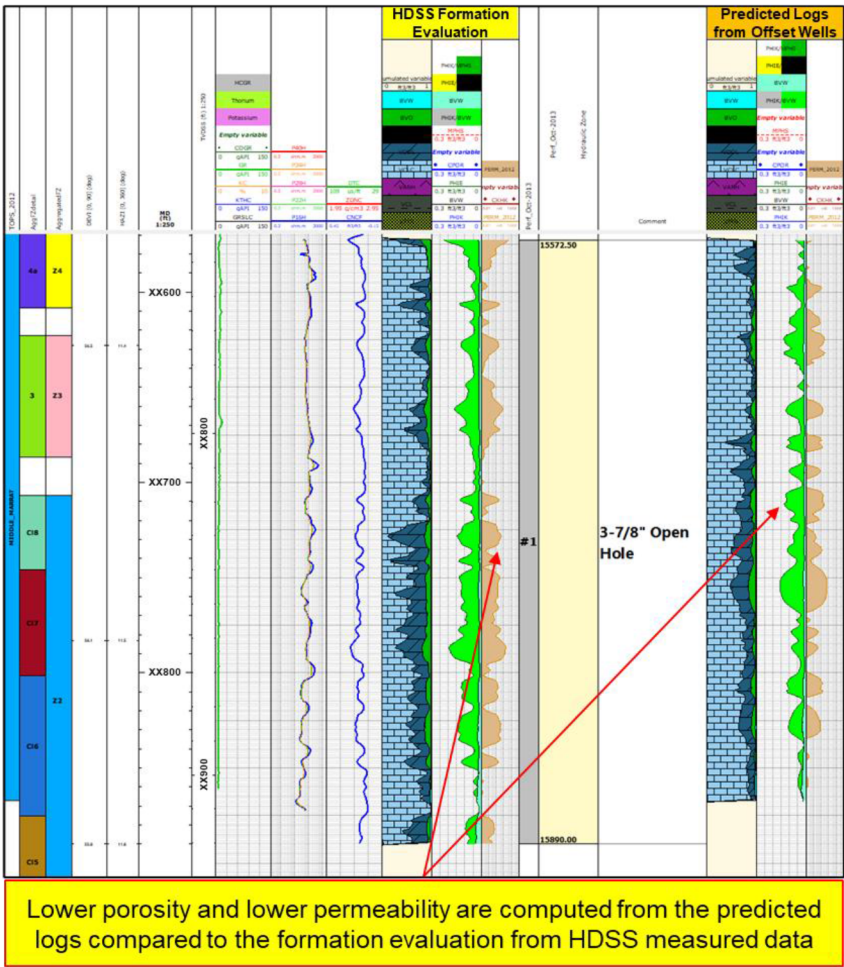
Figure 13—Well-1 Formation Evaluation, Barite Damage & Pin-Point Stimulation Depths

Based on the Barite contamination severity, proper mitigation actions were taken for the pin-point stimulation as tabulated in Table 2. Filter-cake Breaker fluid and High-Viscous (Hi-vis) pills were used to dissolve the filter-cake, clear and circulate it out with the stagnant Barite column. HPAJT's spacing and intensity (Track 9, Figure 13) was determined to establish contact with virgin rock. A highly retarded single-phase acid fluid system was used for stimulation to maximize acid coverage and penetration inside the rock formation.

Table 2—Well-1 Barite Invasion Observations and High-Pressure Abrasive Jetting Tool (HPAJT) Pin-Point Stimulation Planning

Well-1 AbrasiJET Pin-Point Stimulation Staging				
From	To	Zone	Observations	Comments
ft	ft			
XX570	XX610	3	Low barite invasion in low porosity rock	Low density stim
XX630	XX660	3	Low barite invasion in low porosity rock	Low density stim
XX715	XX730	2	Moderate barite invasion	Medium density stim
XX740	XX750	2	Moderate barite invasion	Medium density stim
XX760	XX785	1	Barite settling in the well with barite invasion also	Aggressive stim density
XX800	XX898	1	Barite settling in the well with barite invasion also	Aggressive stim density

The formation evaluation results from the measured HDSS log data reveal better pay characteristics as compared to the predicted logs from offset wells [Figure 14](#).



Well	Dataset	HCPOR-TH (ft)	K.H (mD.ft)
Well-1	Formation Evaluation from HDSS Log Data	22.251	416.556
Well-1	Formation Evaluation from predicted logs	16.131	173.716
HDSS Formation Evaluation yields higher HCPV & K.H		38% Higher	140% Higher

Figure 14—Comparison of Well-1 Computed and Measured Porosity/ Permeability

Treatment Execution

Based on the evaluation of barite invasion (table 2) the pin-point pumping schedule was finalized with the following points:

- Zone 1 –Aggressive stimulation density: where every 10ft HPAJT was stationed for 18 min while jetting stimulation treatment (total of 14 stations)
- Zone 2 – Medium density stimulation: due to the short length of the interval (35 ft) another 3 stations took place
- Zone 3 –Low density stimulation: 9 stations were distributed over the 90 ft of the openhole interval

The execution summary of pin-point stimulation run is shown in Figure 15 and Table 3.

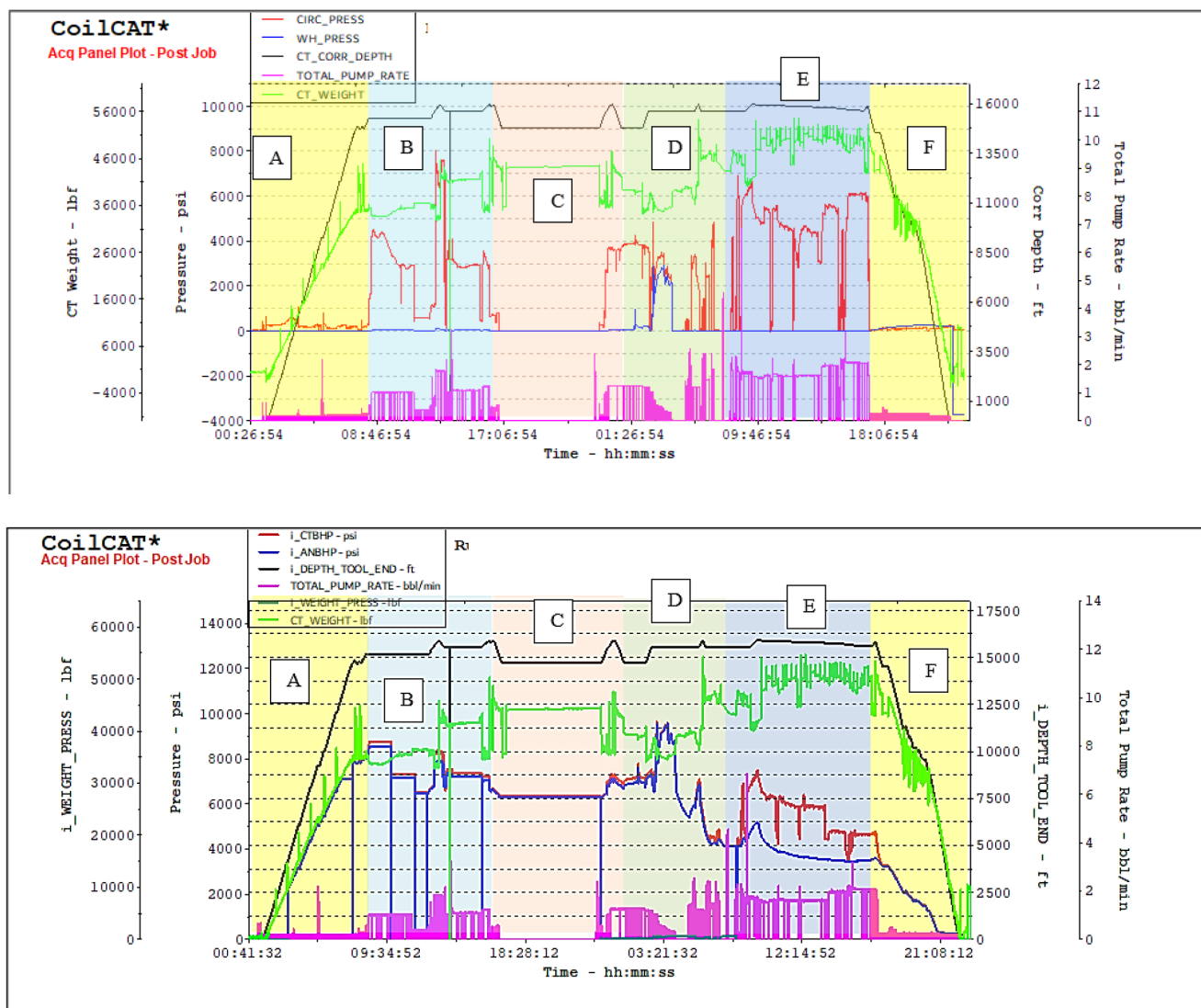


Figure 15—Well-1 Pin-Point-Stimulation Acid Treatment Plot (surface parameters up, downhole parameters down)

Table 3—Operational steps during CT run

Zone	Notes
A	HFOCT RIH with HPAJT while pumping 11.6ppg CaCl ₂ Brine at min rate, depth correlation was performed with CCL signature. Continue RIH to Liner Shoe while pumping 11.6ppg CaCl ₂ Brine at min rate.
B	CT kept CT Stationary displacing 11.6ppg CaCl₂ Brine by pumping 8.4ppg KCl Brine till clear returns, then, resume RIH till EOL at 15,565ft while pumping 8.4ppg KCl Brine at 0.6bpm. Pump rate was increased to 1.8bpm and CT continue RIH till TD while circulating 10bbl HiVIS Brine till Nozzle followed by 8.4ppg KCl Brine. With HiVis at nozzle, start sweeping at 10ftm to liner section while pumping Bottoms Up. CT kept stationary at EOL at 15,565ft MD pumping 8.4ppg KCl Brine till HiVis Brine was circulated out and clear returns observed. Resume RIH till TD at 15,890ft MD while circulating till nozzle 5 bbl HiVis Brine, 8 bbl Filtercake Breaker and 5 bbl HiVis Brine, followed by 8.4 ppg KCl Brine
C	Stop pumping and POOH to EOT for Soaking Time under Shut-in Wellbore conditions. CT resume RIH will TD while pumping 8.4ppg KCl Brine at min rate to circulate out the Filtercake breaker, once confirmed by testing, CT POOH while pumping 8.4ppg KCl Brine till liner section.
D	Start Circulating 15bbl 15% HCl to pickle the tubing displaced by 8.4ppg Brine. CT kept stationary at 14,694ft while Bottoms Up with 8.4ppg KCl Brine, to circulate out Acid and Mud Breaker (Clear returns at Surface). Mixing for the Pin-Point Stimulation. Resume to RIH till EOL at 15565ft MD while circulating 10bbl 15% HCl for injectivity improvement, close the well and perform injectivity test. Spotting 20bbl SPRIAS f in the Open Hole Section while squeezing the remaining of SPRIAS from EOL.
E	Drop 7/16in Ball and chase it with 8.4ppg KCl Brine at 1.8bpm to activate the HPAJT gun, indication observed on differential pressure. Started circulating all the treatment till nozzle, with the well under shut-in conditions started squeezing 78 bbl Pre Flush & Post Flush, 614 bbl SPRIAS and 125 bbl Viscoelastic Diverter. Pin-Point Stimulation was successfully performed, treatment was displaced with 8.4ppg KCl Brine.
F	CT POOH to surface under shut-in wellbore conditions while pumping 8.4ppg KCl Brine.

Operational learnings

The Stage 1 results led to the recommendation to increase near-wellbore diverter volume and concentration in later stages to force improved distribution.

Results and Achievements Summary

With the identification of zones with formation damage and live hydrocarbon zones, Pin-point stimulation conducted for 27 AbrasiJET stations. Well was then active after brief Nitrogen (N₂) lift till 6000 ft. The production test result post stimulation and activation demonstrated the achievement of high Initial Production Results that exceeded the production forecast. The conductive integrated multi-team collaboration on deployed intervention for formation Evaluation & pin-point Stimulation in slim 3-7/8" OH led to achieving:

- First Worldwide Slim Pulse Neutron (High-Definition Spectroscopy) deployed on Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform executed in July 2024.
- Then, First Worldwide Slim Pulse Neutron (High-Definition Spectroscopy) deployed on Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform and Pin-point High Pressure AbrasiJET Stimulation in Well-1 on Rig (October 2024).
 - Formation evaluation using Slim Pulse Neutron logging tool enabled proper formation potential evaluation which led efficient stimulation planning.
 - Detection of Barite invasion in OH via Slim Pulse Neutron logging tool technology flagged on-spot decision of Hi Vis pill circulation
 - Tailored 27 AbrasiJET pin-point stimulation stations unlocked the 3-7/8" slim OH production potential through efficient acid distribution including low perm intervals from Logs.
 - Well-1 achieved results exceeding the initial pre-stimulation expectation (1.5 folds).

Conclusion

- By integrating the technologies bundle as a complete integrated solution, successful safe accessing intervention of the challenging 3-7/8" OH completions for formation evaluation and stimulation is made possible.
- Successful First Worldwide Slim Pulsed Neutron High-Definition Logging & AbrasiJET Pinpoint Stim intervention in 3-7/8" slim OH deployed on Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform.
- Achieved high initial production results exceed the initial production forecast.
- Plan to apply same best practices for 3-7/8"OH in:
 - New drilling wells
 - In existing wells: Review remaining barefoot wells (24×3-7/8" + 3×6") for potential rig-less Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform for Formation Evaluation Logging and AbrasiJet pin-point acid stimulation
 - Workover for cleaning 3-7/8" OH as demanded followed by AbrasiFRAC pin-point acid stimulation
- Solution of Real-time Hybrid Fiber Optics Coiled-tubing Precision Platform based AbrasiJet paved the road for further successful implementations of the approach on drill work-string as integrated scope in rig workover of slim OH cleanout package, for Jurassic OH wells with CTU access restriction.

Nomenclature

HPHT	High Pressure High Temperature
MSC	Multi-Stage Completion
ID	Inner Diameter
PTA	Pressure Transient Analysis
MASAP	Maximum Allowable Surface Pressure
D	Perforation Diameter
GR	Gamma Ray
DLS	Dog log severity
PBR	Polished Bore Receptacle
Kips	1,000 Pounds-wt

OH	Open-hole
KOC	Kuwait Oil Company
H ₂ S	Hydrogen Sulfide
CO ₂	Carbon Dioxide
HPAJT	High-Pressure Abrasive Jetting Tool
CT	Coiled Tubing
HRMA	High Rate Matrix Acidizing
PLT	Production Logging Tool
DEH	Dual External Hydraulic Toe Fracture Sleeve
OD	Outer Diameter
ID	Inner Diameter
TD	Total Depth
OBM	Oil Base Mud
NK	North Kuwait
NKJG	North Kuwait Jurassic Gas
RIH	Running in hole

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